Measuring Learners English Writing - User Guide

A comprehensive linguistic diagnostic tool designed by Oscar Wenzhao He for measuring English learner writing, syntactic complexity, logical connectors, sentence length variation, lexical density, target wordlist marking, indicator statistics, and multi-format CSV / Excel data exports. Follow this guide to get the most out of your linguistic analysis.

Target Users

This tool is meticulously crafted for professionals, researchers, teachers, and learners involved in English language education, linguistics, and text analysis:

• English Language Teachers & Educators (EFL/ESL): Evaluate student essay complexity, assess mastery of academic transition words, and track syllabus vocabulary usage against target wordlists.
• Linguistics & Applied Linguistics Researchers: Conduct corpus-based studies, second-language (L2) syntactic maturity benchmarks, discourse transition frequencies, and statistical indicator distributions across student cohorts.
• Students & English Learners: Receive automated, actionable diagnostics on sentence variety, connector diversity, and target vocabulary coverage.
• Curriculum & Content Developers: Audit instructional materials, measure reading level difficulty, and verify target vocabulary recurrence across textbook units.

When to Use This Tool

"Measuring Learners English Writing" shines in scenarios where deep textual understanding, learner diagnostics, and statistical modeling are required:

• Descriptive Statistics of Writing Indicators: Compute rigorous descriptive statistics (Mean, Standard Deviation, Median, Min, Max, and Total Sum) across dozens of linguistic and syntactic indicators for a corpus of learner essays.
• Direct CSV Data Export for Research: Export publication-ready CSV datasets formatted for immediate import into statistical packages such as R, SPSS, Stata, JASP, Jamovi, or Microsoft Excel.
• Target Wordlist Marking & Coverage Assessment: Supply custom vocabulary lists or choose from built-in academic wordlists (e.g., Coxhead's Academic Word List). The tool automatically marks target words in the text and audits usage rates.
• Syntactic Complexity Evaluation: Assess how intricately a text is structured by calculating coordinate/subordinate clauses, subordination ratios, and complex sentence percentages.
• Lexical Richness Check: Determine if a text utilizes a varied vocabulary (Type-Token Ratio) and a high density of information-carrying content words (Lexical Density).
• Comparative Corpus Studies: Use batch analysis to process multiple documents simultaneously to identify patterns across different authors, student cohorts, or revision drafts.
• Data Cleaning & Stopword Filtering: Toggle stopword filtering to isolate meaningful lexical words from grammatical function tokens.

Statistics of Writing Indicators

The tool performs automated statistical aggregation across all analyzed documents:

• Statistical Metrics Computed:
• Mean (Average): The arithmetic average of the indicator across all documents in the corpus.

• Standard Deviation (SD): Measures the dispersion and variability of the indicator across the learner cohort.
• Median: The central tendency value, robust against outliers and extreme essay lengths.
• Minimum (Min) & Maximum (Max): The observed range across the cohort.
• Total / Sum: The cumulative count across the entire corpus for discrete frequency metrics (words, sentences, clauses, connectors, target vocabulary).
• Indicator Categories Covered:

1. Volume & Length: Total Words, Total Sentences, Mean Sentence Length, Sentence Length SD (Variability), Min/Max/Median Sentence Length, Short/Medium/Long Sentence Ratios.
2. Syntactic Complexity: Clauses Per Sentence, Subordinate Clauses (Count & Ratio), Coordinate Clauses (Count & Ratio), Complex Sentences (Count & Ratio %), Passive Voice Constructions (Count & Ratio).
3. Logical Connectors & Cohesion: Total Logical Connectors, Connector Density per 100 Words, Contrast & Concession, Addition & Elaboration, Cause & Effect, Sequence & Progression, Exemplification & Clarification.
4. Lexical Diversity & Sophistication: Unique Word Types, Type-Token Ratio (TTR), Lexical Density (% content words), Advanced/Polysyllabic Words (>=3 syllables Count & Ratio %), Mean Word Length.
5. Target Wordlist Diagnostics: Total Matches (Tokens), Wordlist Density %, Unique Types Covered, List Coverage Rate %.

• Interactive Indicator Statistics Matrix: Switch to the Indicator Statistics tab in the Batch Analyzer to view the interactive statistics table, complete with category filter pills and key cohort benchmark cards.

Target Wordlist Marking & Curriculum Evaluation

A key capability of this tool is tracking and marking learner uptake of designated target vocabulary:

• Flexible Wordlist Configuration:
• Curated Academic Presets: Instant one-click loading of the Academic Word List (AWL Core), Academic Discourse & Transitions, and Critical & Analytical Verbs.
• Custom Vocabulary Input: Paste or type custom vocabulary sets separated by commas, spaces, semicolons, or newlines.
• File Import: Upload existing `.docx`, `.txt`, or `.csv` wordlists directly.
• Inflection-Aware Matching: Matches base lemmas as well as regular morphological inflections (e.g., analyze, analyzes, analyzed, analyzing).
• In-Text Visual Marking:
• Switch to Marked Text view in the source panel to see all occurrences of target vocabulary highlighted in distinct amber tags with match counts.
• Diagnostic Metrics:
• Target Matches (Tokens): Total occurrences of target words in the essay.
• Target Word Density (%): Proportion of target words relative to total essay word count.
• Types Covered: Number of unique target words from the list utilized by the student.
• List Coverage Rate (%): Percentage of target curriculum vocabulary successfully incorporated.
• Unmatched Words Inventory: Identifies target vocabulary words not yet utilized in the draft—ideal for instructor revision prompts.

Supported Document Formats

The tool natively supports seamless importing of learner texts:

• Microsoft Word Documents (`.docx`): Automatically extracts full essay content without requiring manual conversion or copy-pasting.
• Plain Text Files (`.txt`): Standard UTF-8 text documents.
• Drag-and-Drop: Drag `.docx` or `.txt` files directly into the single editor or the batch dropzone.

CSV & Excel Export Options

All indicators and statistical aggregations can be exported with a single click:

• Batch Analysis CSV Exports:

1. Indicator Statistics Summary (`Corpus_Indicator_Statistics_Summary.csv`): Clean, publication-ready table where rows are indicators and columns are Category, Indicator Name, Unit, N, Mean, SD, Median, Min, Max, and Total Sum.

2. All Documents & Statistics (`Corpus_Indicators_And_Statistics.csv`): Comprehensive dataset with every document as a row, all indicator values, and statistical summary rows (Mean, SD, Median, Min, Max, Sum) appended at the bottom.

3. Target Wordlist Breakdown (`Corpus_Target_Wordlist_Breakdown.csv`): Document-by-document table containing every target word's presence status, frequency count, density %, matched token forms, and sample sentence contexts.

• Single Essay CSV Export:

• Indicators & Statistics CSV (`[Document]_indicators.csv`): Detailed CSV of all measured indicators with units and values, along with target wordlist diagnostic hits.

• Excel Multi-Sheet Workbooks (`.xlsx`):

• Consolidated spreadsheets containing Corpus Stats, Indicator Statistics Summary, Target Wordlist Breakdown, and Word Frequency tables.

Application Menus & Navigation (•oNöƒÜSUN [ü,*|û~ß)

The interface features an intuitive menu bar at the top:

• File Menu (e‡NöƒÜSU):

• Upload Files (Single/Batch): Seamlessly load individual `.docx` or `.txt` essays into the Single Document Analyzer, or batch upload entire corpora into the Batch Corpus Analyzer.

• Export (CSV/Excel): Export publication-ready multi-sheet Excel workbooks (`.xlsx`) or statistical CSV datasets (`.csv`) for individual essays or whole corpora.

• Clear All: Cleanly reset the active document or corpus queue with one click.

• References Menu (SÂ€ e‡s.N ‹¾•nƒÜSU):

• Target Wordlist: Configure curriculum vocabulary lists, select academic presets (AWL Core, Discourse, Verbs), or import external wordlists.

• Filter Stopwords: Globally toggle function/grammar word exclusion on or off across lexical frequency tables and analytical metrics.

• User Guide: Open the complete PDF user guide and methodology documentation.

Preparation & Workflow

1. Prepare Files: Organize student essays in `.docx` (Microsoft Word) or `.txt` (Plain Text) format for single document evaluation or batch corpus upload.

2. Upload via File Menu: Click File -> Upload Files to import your single document or batch corpus. Drag & drop is also natively supported.

3. Configure References: Click References -> Target Wordlist to set active target vocabulary, and toggle References -> Filter Stopwords as needed.

4. Analyze & Inspect: Review full-width indicator profiles, sentence complexity, clause breakdowns, and target wordlist density. Click View Document & Marked Words to view in-text vocabulary highlights.

5. Indicator Statistics: Inspect descriptive statistics (Mean, Median, SD, Min, Max, Sum) for all writing indicators across individual documents or whole corpora.

6. Export via File Menu: Click File -> Export (CSV/Excel) to download multi-sheet Excel workbooks (`.xlsx`) or statistical CSV datasets.

Windows Desktop EXE Compilation & Offline Usage (• ‹ÑN: Windows hL—brH

The software is engineered with a modular Electron 42 and Vite architecture, allowing researchers and teachers to compile and run it as an independent Windows executable (`.exe`):

• 100% Offline & Private: Learner writings, exams, and corpus files are processed completely in memory and on local disk without transmission over external networks.

• One-Click Build Script: Double-click `build-windows-exe.bat` on any Windows PC with Node.js installed to automatically build the application.

• Compilation Commands:

1. `npm install` - Install dependencies.
2. `npm run build:exe` - Builds production React bundle and creates standard Windows installer (`.exe`) in `/release/`.
3. `npm run build:exe:dir` - Produces portable unpacked application directory (`release/win-unpacked/Measuring Learners English Writing.exe`).

• In-App Assistant: Click Windows EXE in the top navigation bar to download the batch script and view step-by-step compilation guidance.

Designed by Oscar Wenzhao He